

Business Math I (MT1002)

Sessional-II Exam

Date: November 3rd 2025

Course Instructor(s)

Dr. Hina Firdous, Mr. Saleem Chishti, Dr. Aziz, Dr. Amir Hussain, Dr. Kanwal, Dr. Zoha, Ms Maheen Sultan, Mr. Abid Ali

Total Time : 1 Hour

Total Marks: 30

Total Questions: 3

25L-5575

Roll No

BSCEU

Section

Shayan

Student Signature

CLO # 2: To learn Exponential Growth and decay functions and its importance

Q.No.1:

(a) Solve the following equation:

$$2 \log(5x) - \log(2x) = 2.5 \quad [4]$$

(b) A consumption function is modeled by the equation

$$C = 500(1 - e^{-0.3x})$$

Where x is income given in thousands of dollar

Graph the consumption function over the interval $0 \leq x \leq 50$. What value does consumption approach if income increases without limit. Use the graph to verify your answer.

[2+3+1]

CLO # 3: Financial Mathematics problems.

Q.No.2:

(a) If a savings account awards interest of 6% per year compounded quarterly, what amount must be deposited today in order to accumulate \$25,000 after 8 years? How much interest will be earned?

[5+1]

(b) The manufacturer of a new cholesterol-reducing yogurt plans to increase output over the coming years. The present output is 5500 units per week. If the output is increased by 150 per week, calculate the number of weeks before the weekly output exceeds 10000.

[4]

CLO # 5: To discuss the techniques of Differentiation & Integration and its applications for business students

Q.No.3:

(a) Sketch the curve $f(x) = x^3 + 9x^2 + 27x + 27$ by determining the turning points, increasing/decreasing intervals, and maxima/minima.

[2+3+1]

(b) The demand function for a monopolist is $Q = 120 - 3P$. Evaluate TR, MR and AR at $Q = 15$. Hence explain in words the meaning of each function at $Q = 15$.

[4]